

Inventory of inputs, with sources

Compiled for the `ionization_yield / photon_vs_electron` project

2026-09-14

Every number this project takes from outside itself, with its symbol, value, range, units and source. **Values are read from the code that uses them** by `make_inputs_table.py`, so the table cannot drift from the calculation; only the symbols, units and citations are written by hand.

What the status column is for. It separates four very different kinds of number that a bare citation would make look alike: a value checked numerically against the paper it came from; a value whose paper we have never opened; a value you chose; and a value that is conventional in the field but for which *this project has established no source at all*. The last category is the one worth acting on.

1 Inventory of inputs

Every externally supplied number in this project. Values are read from the code that uses them, so this table cannot drift from the calculation. Status: **V** verified numerically against the source, which is in `papers/`; **C** cited but the source is not in the tree and the implementation is unchecked; **D** derived here; **S** scanned; **U** your stated choice; **X** conventional value whose source this project has *not* established.

symbol	value	range	units	source	st.
Cosmology					
h	0.6766	—	—	[26]	C
Ω_m	0.3111	—	—	[26]	C
Ω_Λ	0.6889	—	—	[26]	D
flat by assumption					
$\Omega_b h^2$	0.02242	—	—	[26]	C
ρ_{crit}/h^2	1.87834e-29	—	g cm^{-3}	[31]	C
$T_{\text{CMB},0}$	2.7255	—	K	[6]	C
Y_P	0.245	—	—	[26]	C
BBN + Planck					
X_H	0.755	—	—	[26]	D
$1 - Y_P$					
$H(z)$	Planck18	—	s^{-1}	[26]	C
astropy; includes radiation					
IGM target					
$n_{\text{H}}(z=10)$	2.5287e-04	—	cm^{-3}	[26]	D
proper; from $\Omega_b h^2$, X_H					
$N_{\text{HI,norm}}$	1759.467095	—	m^{-3}	[26]	D
$n_{\text{HI}}(z) = N((1+z)/21)^3$; harmonised					
x_e	1e-04	—	—	—	U
held static; your value					
B_0	1.0e-13	—	T	—	U

symbol	value	range	units	source	st.
1 nG comoving; $B \propto (1+z)^2$					
Z	1	—	—	—	U
pure hydrogen target					
δ_{Gould}	0.5	—	—	[8]	C
max fractional transfer					
f_{therm}	1.5	—	—	—	U
$K_{\text{term}} = f k T_{\text{CMB}}$					
$n_{\text{max}}^{\text{exc}}$	10	—	—	[28]	C
levels $n = 2 \dots n_{\text{max}}$					
Atomic physics					
$E_{\text{th}}(\text{HI})$	13.598434600	—	eV	[16]	V
R_{∞}	13.605693123	—	eV	[30]	V
B (Kim Table I)	13.6057	—	eV	[14]	V
the Rydberg; tension with NIST at 0.05%					
U/B	1	—	—	[14]	V
virial, H(1s)					
N_i	0.434306	—	—	[14]	V
continuum oscillator strength					
M^2	0.2834	—	—	[14]	V
a_i (df/dw)	4 coeffs	—	—	[14]	V
Table I, H(1s)					
σ_{ion}	RBED eq. (20)	—	m ²	[15]	V
verified to 1.4×10^{-9}					
dσ/dw	BED, non-rel.	—	m ² eV ⁻¹	[14]	V
relativistic factor omitted: +25% at 1 MeV					
σ_{exc}	Stone & Kim	—	m ²	[28]	C
Coulomb loss	Gould (1972)	—	eV s ⁻¹	[8]	C
IC / brems	BG70	—	eV s ⁻¹	[2]	C
eq. (2.42) spectrum verified; loss rates not					
I_{exc}	14.99	—	eV	[11]	C
mean excitation energy, route 1 only					
$\alpha_B(2 \times 10^4 \text{ K})$	1.430e-13	—	cm ³ s ⁻¹	[23]	V
confirmed vs [10] to 0.16%					
χ	1.08	—	—	[20]	V
derived here as $1 + (Y_P/4)/X_H = 1.0811$					
Stellar source					
log ₁₀ ρ _{UV}	25.12	+0.07/ − 0.08	erg s ⁻¹ Hz ⁻¹ cMpc ⁻³	[5]	V
Table 3, $z = 10$ row					
log ₁₀ ξ _{ion}	25.28	±0.42 dex	Hz erg ⁻¹	[19]	V
median at $z \simeq 7.14$; scatter is observed, not error					
K_{UV}	1.150e-28	—	$M_{\odot}\text{yr}^{-1}/(\text{erg s}^{-1}\text{Hz}^{-1})$	[20]	V
eq. (10); Salpeter 0.1–100 $M_{\odot}$					
α_{IMF}	2.35	—	—	[27]	V
fitted for 0.4–10 $M_{\odot}$ only					
$m_{\text{min}}, m_{\text{max}}$	0.1, 100.0	—	$M_{\odot}$	[20]	C
range adopted to match K_{UV}					
m_{SN}	8.0	—	$M_{\odot}$	—	U
core-collapse threshold					

symbol	value	range	units	source	st.
SN per $M_\odot$	7.4218e-03	—	$M_\odot^{-1}$	[27]	D
IMF integral, this work					
α_{SED}	2	1–3	—	[21]	C
$f_\nu \propto \nu^{-\alpha}$; BPASS spectrum unobtainable					
LyC band	1–4	—	Ryd	—	U
conventional					
T_{eff}	5e+04	—	K	—	U
blackbody, comparison case only					
Cosmic-ray channel					
E_{SN}	1e+51	—	erg	—	X
standard core-collapse budget; source not established here					
ϵ_{CR}	0.1	0.1–0.2	—	[3]	C
hybrid simulations: 10–20% of bulk kinetic energy into non-thermal protons at parallel/quasi-parallel strong shocks, dropping to ZERO when quasi-perpendicular; $\simeq 10\%$ inferred for Tycho. An obliquity-AVERAGED value would be lower, so 0.1 favours the CR channel					
$f_e = K_{ep}$	0.01	$1-3 \times 10^{-3}$ (source)	—	[24]	C
the project uses the value measured at Earth, not the value the source derives: [24] give $K_{ep} \simeq 1-3 \times 10^{-3}$ at realistic SNR shock speeds and 1.6×10^{-3} for Tycho, so 0.01 is 3–10 $\times$ high and favours the CR channel. K_{ep} is defined at fixed MOMENTUM and is independent of p					
p	2.2	—	—	—	U
your choice; DSA strong-shock predicts 2					
$E_{\text{min}}, E_{\text{max}}$	1e+03, 1e+12	—	eV	—	U
your choice; spectrum cancels from ζ_e					
Reionization budget					
$C_{\text{IGM}}(z)$	$1 + 43z^{-1.71}$	—	—	[25]	V
eq. (A1), C_{100} , r19.5 run; $43 = e^{3.76}$; fit valid $6 \leq z \leq 20$					
$\langle t_{\text{rec}} \rangle$	eq. (24)	—	Gyr	[20]	V
3.2 Gyr normalisation reproduced to 1.6%					
Scanned ranges					
f_{esc}	0.1	10^{-6} –1	—	—	S
canonical value is a convention, not a measurement					
$\epsilon_{\text{CR}} f_e$	1e-03	10^{-6} –1	—	—	S
bounded above by unity on energy grounds					
ρ_{SFR} scale	1	10^{-3} –1	—	—	S
ignorance band at $z = 20$ only, where ρ_{UV} is unmeasured					
Source spectra					
α_{EUV} (RQ)	1.57	± 0.17	—	[29]	V
HST composite, rest-frame 500–1200 Å					
α_{EUV} (RL)	1.96	± 0.12	—	[29]	V
radio-loud quasars; full composite 1.76 ± 0.12					
α_{AGN}	1.5	—	—	[9]	V
EoR quasar template; corroborates [29]					
AGN band top	3	—	keV	[9]	V
13.6 eV–3 keV; above 4 Ryd is an extrapolation					
Γ_1 (ULX)	2.1	1.38–3.1	—	[7]	V
Table 6; single-PO fits $1.6 < \Gamma < 3.3$					
E_{break} (ULX)	5.0	3.5–7	keV	[7]	V
broken PL preferred > 98% in 11 of 12; fits are 2–10 keV only					
$\Delta\Gamma$ (ULX)	1.5	1–2	—	[7]	V

symbol	value	range	units	source	st.
steepening above the break					
kT_e, τ (ULX corona)	1–3, 6–80	–	keV, –	[7]	V
cool, optically THICK; global χ^2 min in 10/12					
Γ (BHB hard)	1.7	< 2.1	–	[7]	X
the typical $\Gamma \simeq 1.7$ is second-hand; only $\Gamma < 2.1$ for the low/hard state is first-hand (§4.1)					
p per class	2.2	–	–	—	U
held at the project fiducial: no citable per-class CR index at $z \sim 10$					
Source luminosities					
AGN LyC fraction	0.17–0.33	–	–	[1]	V
17% ([1]) to $\leq 1/3$ ([13]), both at $z \sim 7.5$; carried to $z = 10$, conservative for photons					
L_{2-10}/SFR	1.8e+39	1.4e+39–2.3e+39	erg s ^{−1} /($M_\odot\text{yr}^{-1}$)	[18]	V
eq. (6), $\times(1+z)^1$; exponent fixed at 1					
L_{mech} (microquasar)	1e+39–1e+40	–	erg s ^{−1}	[22]	V
SS 433 and the S26 microquasar; TOTAL injected, not the electron share					
E_{mech} per object	1e+54	–	erg	[22]	V
whole lifetime; $\gg$ a core-collapse SN					
M_{UV} (U37126)	−20.1	± 0.05	AB	[21]	V
$z = 10.255$, lensed $\mu \simeq 2.2$; the per-object stellar anchor					
$f_{\text{jet,e}}$	1e−03–1	scanned	–	—	S
jet efficiency $\times$ leptonic fraction, scanned: the proton content of jets is not observationally constrained					
f_{dep}	1	–	–	—	U
assumption A11: every injected electron deposits — a CEILING					
Derived (this work)					
W_e	48.038	–	eV ion ^{−1}	—	D
$\langle E \rangle / \langle N \rangle$ of the fiducial CR spectrum					
W_γ (stellar)	21.07	–	eV ion ^{−1}	—	D
$W_\gamma/W_e = 0.439$; IGM transparency applied					
W_γ (AGN)	27.41	–	eV ion ^{−1}	—	D
$W_\gamma/W_e = 0.571$; IGM transparency applied					
W_γ (ULX)	38.78	–	eV ion ^{−1}	—	D
$W_\gamma/W_e = 0.807$; IGM transparency applied					
W_γ (BHB hard)	146.83	–	eV ion ^{−1}	—	D
$W_\gamma/W_e = 3.057$; IGM transparency applied					
$\tau_{\text{IGM}}(1 \text{ keV})$	1.93	–	–	—	D
over one Hubble length at $z = 10$; thick below $\sim 500 \text{ eV}$, transparent above $\sim 3 \text{ keV}$					
$\tau_{\text{IGM}}(3 \text{ keV})$	0.0476	–	–	—	D
over one Hubble length at $z = 10$; thick below $\sim 500 \text{ eV}$, transparent above $\sim 3 \text{ keV}$					
$\tau_{\text{IGM}}(10 \text{ keV})$	0.00077	–	–	—	D
over one Hubble length at $z = 10$; thick below $\sim 500 \text{ eV}$, transparent above $\sim 3 \text{ keV}$					
$t_H(z=10)$	0.7086	–	Gyr	[26]	D
astropy Planck18, complete $H(z)$					

2 The fiducial top-heavy IMF

For the scenario considered here — Population II stars in galaxies at $z = 10$ — the fiducial top-heavy IMF adopted is that of Jeong, Jeon, Song & Bromm [12], their eq. (2):

$$\xi(m) = \frac{dN}{dm} = \xi_0 m^{-\alpha}, \quad \alpha = 1.3, \quad 1 M_\odot \leq m \leq 100 M_\odot. \quad (1)$$

It is chosen over the alternatives for four reasons. *First*, it is stated explicitly as a top-heavy IMF for Pop II stars, not for Pop III, which is the population this project’s galaxies contain. *Second*, the galaxies it is applied to form at $z = 10$, the redshift of this calculation. *Third*, it is the same functional form as the Salpeter law already in use, $dN/dm \propto m^{-\alpha}$, so it substitutes directly into the supernova-rate integral with no change of machinery — only α and $m_{\min}$ move. *Fourth*, the physical motivation is the one relevant here: a raised temperature floor from low-metallicity gas radiatively coupled to a CMB at $T = 2.73(1+z)$ K, which lifts the Jeans mass and hence the characteristic stellar mass [17]. Their default Pop II IMF, for comparison, is Chabrier [4] over $0.1\text{--}100 M_{\odot}$.

Not yet adopted. Equation (1) is recorded here as the fiducial to test against; it has *not* been substituted into any result in this project. Doing so would move three of the entries in §1 at once — α_{IMF} , the mass range, and the supernova rate per $M_{\odot}$ — and would also invalidate K_{UV} , which [20] quote for a Salpeter IMF over $0.1\text{--}100 M_{\odot}$. Those three quantities are not independent, and that coupling is the open question this inventory was built to expose.

References

- [1] Shikhar Asthana, Martin G. Haehnelt, Girish Kulkarni, James S. Bolton, Prakash Gaikwad, Laura C. Keating, and Ewald Puchwein. The impact of faint AGN discovered by JWST on reionization. *Monthly Notices of the Royal Astronomical Society*, 2024. VERIFIED from the abstract page. QSO-assisted model: AGN supply 17% of the total hydrogen-ionizing emissivity with 1% of haloes hosting AGN; a QSO-only model needs 10% of haloes hosting AGN. “A modest AGN contribution to reionization aligns with the Lyman- α forest data.” Lower end of the AGN box. AGREES with [13]: both place AGN in the 17–33% minority band. A widely-circulated “36–88%” figure comes from a separate broad-line census with much larger systematics and is NOT supported by either of these.
- [2] George R. Blumenthal and Robert J. Gould. Bremsstrahlung, synchrotron radiation, and Compton scattering of high-energy electrons traversing dilute gases. *Rev. Mod. Phys.*, 42(2):237–270, 1970.
- [3] Damiano Caprioli and Anatoly Spitkovsky. Simulations of ion acceleration at non-relativistic shocks. I. acceleration efficiency. *The Astrophysical Journal*, 783:91, 2014. IN TREE. Hybrid simulations: 10–20% of the bulk kinetic energy goes into non-thermal protons at parallel and quasi-parallel strong shocks, dropping to ZERO for quasi-perpendicular ones. About 10% inferred for Tycho (Morlino & Caprioli 2012). The source for ϵ_{CR} .
- [4] Gilles Chabrier. Galactic stellar and substellar initial mass function. *Publications of the Astronomical Society of the Pacific*, 115:763–795, 2003.
- [5] C. T. Donnan et al. JWST PRIMER: A new multi-field determination of the evolving galaxy UV luminosity function. *Monthly Notices of the Royal Astronomical Society*, 533:3222, 2024. IN TREE. Table 3, $z = 10$ row: $\log \rho_{\text{UV}} = 25.12^{+0.07}_{-0.08}$, integrated down to $M_{\text{UV}} = -17$. They convert with $K_{\text{UV}} = 1.15 \times 10^{-28}$ (Salpeter), the same value used here.
- [6] D. J. Fixsen. The temperature of the cosmic microwave background. *The Astrophysical Journal*, 707:916–920, 2009.
- [7] Jeanette C. Gladstone, Timothy P. Roberts, and Chris Done. The ultraluminous state. *Monthly Notices of the Royal Astronomical Society*, 397(4):1836–1851, 2009. IN TREE (added 2026-09-14). VERIFIED from the PDF, Table 6 and §4.3: a broken power law is preferred over a single power law at $> 98\%$ in 11 of 12 ULXs, with break energies 3.5–7 keV and steepening $\Delta\Gamma \simeq 1\text{--}2$; $\Gamma_1 \simeq 1.38\text{--}3.1$ (typical 2.1), single-PO fits $1.6 < \Gamma < 3.3$ (§4.1). Preferred physical model (§4.5.1, Table 8): a COOL, OPTICALLY THICK corona $kT_e \sim 1\text{--}3$ keV, $\tau \sim 6\text{--}80$

- (global χ^2 minimum in 10/12), against a hot thin corona $kT_e \sim 50$ keV, $\tau \lesssim 1$ as only a local minimum. The source for the ULX/super-Eddington photon channel. NOTE the fits are over 2–10 keV only; the 13.6 eV–2 keV ionizing band is NOT constrained by this paper and is an extrapolation, where the cool disc ($kT_{\text{in}} \sim 0.17$ –1.7 keV, Table 5) actually dominates. NOTE also [22] paraphrases the break as “above 3 keV”; the fitted range is 3.5–7 keV.
- [8] Robert J. Gould. Energy loss of fast electrons and positrons in a plasma. *Physica*, 60(1):145–154, 1972.
 - [9] Luca Graziani, Benedetta Ciardi, and Max Glatzle. X-ray ionization of the intergalactic medium by quasars. *Monthly Notices of the Royal Astronomical Society*, 479(4):4320–4335, 2018. VERIFIED from the article page. Purpose-built reionization-era quasar ionizing SED: a power law $S(\nu) \propto \nu^{-1.5}$ over 13.6 eV–3 keV (soft X-ray band 200 eV–3 keV). Independently corroborates [29]’s radio-quiet $\alpha_{\text{EUV}} = 1.57 \pm 0.17$. The source for the AGN photon channel. Their own footnote 7 flags that a single power law is often inappropriate for sources spanning UV to X-ray — which is why the corona/ULX-like hard case is carried separately.
 - [10] Lam Hui and Nickolay Y. Gnedin. Equation of state of the photoionized intergalactic medium. *Monthly Notices of the Royal Astronomical Society*, 292:27–42, 1997. IN TREE. Case-B $\alpha_B(T)$ fit used to confirm $\alpha_B(2 \times 10^4 \text{ K}) = 1.43 \times 10^{-13}$ to 0.16%.
 - [11] ICRU. Stopping powers for electrons and positrons. Technical Report Report 37, International Commission on Radiation Units and Measurements, 1984. NOT in tree. Mean excitation energy $I = 14.99$ eV; route 1 only.
 - [12] Tae Bong Jeong, Myoungwon Jeon, Hyunmi Song, and Volker Bromm. Simulating high-redshift galaxies: Enhancing UV luminosity with star formation efficiency and a top-heavy IMF. *The Astrophysical Journal*, 2025. IN TREE. eq. (2): top-heavy Pop II IMF, $\xi(m) \propto m^{-1.3}$, 1–100 $M_\odot$, galaxies at $z = 10$.
 - [13] Danyang Jiang, Linhua Jiang, Shengxiu Sun, Weiyang Liu, and Shuqi Fu. AGN ruled out as the dominant source of cosmic reionization. *Nature Astronomy*, 2025. VERIFIED from the abstract page. Separates point-source (AGN) from extended components in JWST FUV imaging down to $M_{\text{UV}} \approx -15$, over $7.15 \leq z \leq 7.75$. Result: AGN supply AT MOST ONE THIRD of the LyC photon budget at $z \sim 7.5$, which rules them out as the dominant ionizing source. Upper bound on the AGN box of the source-competition figure. NOTE measured at $z \sim 7.5$; its use at $z = 10$ is an extrapolation, and a CONSERVATIVE one, since the AGN luminosity function declines faster with redshift than the galaxy one.
 - [14] Yong-Ki Kim and M. Eugene Rudd. Binary-encounter-dipole model for electron-impact ionization. *Phys. Rev. A*, 50(5):3954–3967, 1994.
 - [15] Yong-Ki Kim, José Paulo Santos, and Fernando Parente. Extension of the binary-encounter-dipole model to relativistic incident electrons. *Phys. Rev. A*, 62(5):052710, 2000.
 - [16] A. Kramida, Yu. Ralchenko, J. Reader, and NIST ASD Team. NIST atomic spectra database. National Institute of Standards and Technology. HI ionization energy 13.598434599702 eV.
 - [17] Richard B. Larson. Early star formation and the evolution of the stellar initial mass function in galaxies. *Monthly Notices of the Royal Astronomical Society*, 301:569–581, 1998. NOT in tree. CMB temperature-floor argument for a rising characteristic mass at high z ; cited through [12].
 - [18] B. D. Lehmer, A. R. Basu-Zych, S. Mineo, W. N. Brandt, R. T. Eufrasio, T. Fragos, A. E. Hornschemeier, B. Luo, Y. Q. Xue, F. E. Bauer, M. Gilfanov, P. Lira, J. L. Racusin, D. P. Schneider, O. Shemmer, P. Tzanavaris, F. Vito, M. Yukita, and A. Zezas. The evolution of normal galaxy X-ray emission through cosmic history: Constraints from the 6 Ms Chandra

- deep field-south. *The Astrophysical Journal*, 825(1):7, 2016. VERIFIED from the article page, eq. (6): $L_{2-10\text{ keV}}(\text{HMXB})/\text{SFR} = (1.8_{-0.4}^{+0.5}) \times 10^{39} (1+z) \text{ erg s}^{-1} (\text{M}_{\odot} \text{ yr}^{-1})^{-1}$, band 2–10 keV, redshift exponent exactly 1. The source for the HMXB/microquasar luminosity normalisation. CAUTION: a widely-quoted variant $10^{39.28}(1+z)^{1.31}$ circulates in secondary summaries and is NOT this paper’s eq. (6) — checked and rejected.
- [19] M. Llerena, L. Pentericci, L. Napolitano, et al. The ionizing photon production efficiency of star-forming galaxies at $z \sim 4 - 10$. *Astronomy & Astrophysics*, 698:A302, 2025. IN TREE. Median $\log \xi_{\text{ion}} = 25.28$ at $z \simeq 7.14$, observed scatter 0.42 dex.
 - [20] Piero Madau and Mark Dickinson. Cosmic star-formation history. *Annual Review of Astronomy and Astrophysics*, 52:415–486, 2014. IN TREE. eq. (10) $K_{\text{UV}} = 1.15 \times 10^{-28}$, a compromise over age and metallicity; eq. (24) $\langle t_{\text{rec}} \rangle$.
 - [21] R. Marques-Chaves et al. A study of the $z = 10.255$ galaxy U37126. *Astronomy & Astrophysics*, 2026. IN TREE. BPASS v2.2.1, $Z = 0.003$, 6.8 Myr population; the LyC shape adopted for the stellar SED.
 - [22] I. F. Mirabel, M. Dijkstra, P. Laurent, A. Loeb, and J. R. Pritchard. Stellar black holes at the dawn of the Universe. *A&A*, 528:A149, 2011.
 - [23] Donald E. Osterbrock and Gary J. Ferland. *Astrophysics of Gaseous Nebulae and Active Galactic Nuclei*. University Science Books, 2nd edition, 2006. NOT in tree. Table 2.1: case-B recombination coefficients.
 - [24] Jaehong Park, Damiano Caprioli, and Anatoly Spitkovsky. Simultaneous acceleration of protons and electrons at nonrelativistic quasiparallel collisionless shocks. *Phys. Rev. Lett.*, 114(8):085003, 2015. IN TREE (added 2026-09-14). Defines $K_{ep} \equiv f_e(p)/f_p(p)$ for $p \gtrsim p_{\text{inj}}$ – at the same MOMENTUM. Both species share the universal DSA index p^{-4} , so “ K_{ep} is independent of p ”. $K_{ep} \approx 1-3 \times 10^{-3}$ at realistic SNR shock speeds, $\approx 1.6 \times 10^{-3}$ inferred for Tycho, ≈ 0.01 for CRs measured at Earth. The source for f_e ; note the project uses the Earth value, not the source one.
 - [25] Andreas H. Pawlik, Joop Schaye, and Eveline van Scherpenzeel. Keeping the universe ionized: photoheating and the clumping factor of the high-redshift intergalactic medium. *MNRAS*, 394(4):1812–1824, 2009.
 - [26] Planck Collaboration, N. Aghanim, Y. Akrami, et al. Planck 2018 results. vi. cosmological parameters. *A&A*, 641:A6, 2020.
 - [27] Edwin E. Salpeter. The luminosity function and stellar evolution. *The Astrophysical Journal*, 121:161–167, 1955. IN TREE. Power law quoted for $\log(M/M_{\odot}) \in [-0.4, +1.0]$ only.
 - [28] Philip M. Stone, Yong-Ki Kim, and Jean Paul Desclaux. Electron-impact cross sections for dipole- and spin-allowed excitations of hydrogen, helium, and lithium. *Journal of Research of the National Institute of Standards and Technology*, 107(4):327–337, 2002.
 - [29] Randal C. Telfer, Wei Zheng, Gerard A. Kriss, and Arthur F. Davidsen. The rest-frame extreme-ultraviolet spectral properties of quasi-stellar objects. *The Astrophysical Journal*, 565(2):773–785, 2002. VERIFIED from the abstract page. HST composite over rest-frame 500–1200 Å. With $f_{\nu} \propto \nu^{-\alpha}$: $\alpha_{\text{EUV}} = 1.57 \pm 0.17$ (radio-quiet), 1.96 ± 0.12 (radio-loud), 1.76 ± 0.12 (full composite). The source for the AGN photon index. NOTE the fit does not extend above 24.8 eV; any harder band top is an extrapolation.
 - [30] Eite Tiesinga, Peter J. Mohr, David B. Newell, and Barry N. Taylor. CODATA recommended values of the fundamental physical constants: 2018. *Reviews of Modern Physics*, 93:025010, 2021.

- [31] R. L. Workman et al. Review of particle physics. *Progress of Theoretical and Experimental Physics*, 2022:083C01, 2022. Particle Data Group. Critical-density coefficient $\rho_{\text{crit}}/h^2 = 1.87834 \times 10^{-29} \text{ g cm}^{-3}$.